

Fitting logistic regression models

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Logistic regression model

Let $(\mathbf{x}_1, Y_1), \dots, (\mathbf{x}_n, Y_n)$ be iid samples from the model

$$Y_i | \mathbf{x}_i \sim \text{Bernoulli}(p_i)$$
$$\log \left(\frac{p_i}{1 - p_i} \right) = \mathbf{x}_i^T \boldsymbol{\beta}$$

(where the distribution of $\mathbf{x}_i$ does not depend on $\boldsymbol{\beta}$).

How do we estimate the coefficients $\boldsymbol{\beta}$?

Maximum likelihood for logistic regression

Let $(\mathbf{x}_1, Y_1), \dots, (\mathbf{x}_n, Y_n)$ be iid samples from the model

$$Y_i | \mathbf{x}_i \sim \text{Bernoulli}(p_i)$$
$$\log \left(\frac{p_i}{1-p_i} \right) = \mathbf{x}_i^T \beta \quad p_i = \frac{e^{\mathbf{x}_i^T \beta}}{1 + e^{\mathbf{x}_i^T \beta}}$$

where the distribution of $\mathbf{x}_i$ does not depend on β . $1-p_i = \frac{1}{1+e^{\mathbf{x}_i^T \beta}}$

$$f(Y_i | \mathbf{x}_i, \beta) = p_i^{Y_i} (1-p_i)^{1-Y_i}$$
$$L(\beta | \mathbf{y}, \mathbf{X}) \propto \prod_{i=1}^n f(Y_i | \mathbf{x}_i, \beta) = \prod_{i=1}^n p_i^{Y_i} (1-p_i)^{1-Y_i}$$

$$\begin{aligned} \ell(\beta | \mathbf{y}, \mathbf{X}) &= \log L(\beta | \mathbf{y}, \mathbf{X}) \\ &\stackrel{\text{up to a constant}}{=} \sum_{i=1}^n [Y_i \log(p_i) + (1-Y_i) \log(1-p_i)] \\ &= \sum_i \left[Y_i \log \left(\frac{p_i}{1-p_i} \right) + \log(1-p_i) \right] \\ &= \sum_i \left[Y_i \mathbf{x}_i^T \beta - \log(1 + e^{\mathbf{x}_i^T \beta}) \right] \end{aligned}$$

Maximizing

$$\ell(\beta|\mathbf{y}, \mathbf{X}) \stackrel{\text{(up to a constant)}}{=} \sum_{i=1}^n \left\{ Y_i \mathbf{x}_i^T \beta - \log(1 + e^{\mathbf{x}_i^T \beta}) \right\}$$

Differentiating:

$$\begin{aligned} U(\beta) &= \frac{\partial}{\partial \beta} \ell(\beta|\mathbf{y}, \mathbf{X}) = \\ \frac{\partial}{\partial \beta} Y_i \mathbf{x}_i^T \beta &= Y_i \mathbf{x}_i \quad \frac{\partial}{\partial \beta} \log(1 + e^{\mathbf{x}_i^T \beta}) \\ &= \frac{1}{1 + e^{\mathbf{x}_i^T \beta}} e^{\mathbf{x}_i^T \beta} \cdot \mathbf{x}_i \\ &= p_i \mathbf{x}_i \\ \Rightarrow U(\beta) &= \sum_{i=1}^n \{ Y_i \mathbf{x}_i - p_i \mathbf{x}_i \} = \sum_{i=1}^n (Y_i - p_i) \mathbf{x}_i \\ &= \mathbf{X}^T (\mathbf{Y} - \mathbf{p}) \stackrel{\text{set}}{=} 0 \end{aligned}$$

Newton's method

- ▶ Want β^* such that $U(\beta^*) = \mathbf{0}$
- ▶ Begin with initial estimate $\beta^{(r)}$
- ▶ Iterative updates:

$$\beta^{(r+1)} = \beta^{(r)} - \left(\mathbf{H}(\beta^{(r)}) \right)^{-1} U(\beta^{(r)})$$

$$U(\beta) = \frac{\partial}{\partial \beta} \ell(\beta | \mathbf{y}, \mathbf{X}) = \mathbf{X}^T (\mathbf{y} - \mathbf{p})$$

$$\mathbf{H}(\beta) = \frac{\partial}{\partial \beta} U(\beta) = \frac{\partial}{\partial \beta} \mathbf{X}^T (\mathbf{y} - \mathbf{p}) = -\mathbf{X}^T \mathbf{W} \mathbf{X}$$
$$\mathbf{W} = \text{diag}(p; (1-p))$$

Possible stepping criteria:

- $\|\beta^{(r+1)} - \beta^{(r)}\| < \delta$
- $|\ell(\beta^{(r+1)}) - \ell(\beta^{(r)})| < \delta$

Re-writing the update rule

Newton's method update rule:

$$\beta^{(r+1)} = \beta^{(r)} + (\mathbf{X}^T \mathbf{W}^{(r)} \mathbf{X})^{-1} \mathbf{X}^T (\mathbf{y} - \mathbf{p}^{(r)})$$

HW from 7.11:

$$= (\mathbf{X}^T \mathbf{W}^{(r)} \mathbf{X})^{-1} \mathbf{X}^T \mathbf{W}^{(r)} \left(\underbrace{\mathbf{X} \beta^{(r)}}_{\text{linear piece}} + \underbrace{(\mathbf{W}^{(r)})^{-1} (\mathbf{y} - \mathbf{p}^{(r)})}_{\text{working residuals}} \right)$$

$$= (\mathbf{X}^T \mathbf{W}^{(r)} \mathbf{X})^{-1} \mathbf{X}^T \mathbf{W}^{(r)} \underset{\substack{\uparrow \text{working} \\ \text{response}}}{\mathbf{z}^{(r)}}$$

iteratively reweighted least squares (IRLS)

Important bc a lot of tools from (weighted) linear regression can then be applied to GLMs! (hat matrix, leverage, etc.)

Weighted least squares

Mat matrix for logistic reg:
 $W^{\frac{1}{2}} X (X^T W X)^{-1} X^T W^{\frac{1}{2}}$
 $W = \text{diag}(\hat{p}_i(1-\hat{p}_i))$

Suppose that we had

$$y = X\beta + \varepsilon, \quad \varepsilon \sim N(0, \Sigma)$$

Let $W = \Sigma^{-1}$ (inverse variance weighting)

$$W^{\frac{1}{2}} \varepsilon \sim N(0, W^{\frac{1}{2}} \Sigma (W^{\frac{1}{2}})^T)$$

$$= N(0, \Sigma^{-\frac{1}{2}} \Sigma \Sigma^{-\frac{1}{2}}) = N(0, I)$$

$\Rightarrow$ now can use OLS to estimate β !

$$W^{\frac{1}{2}} y = W^{\frac{1}{2}} X \beta + W^{\frac{1}{2}} \varepsilon$$

$$y_w = X_w \beta + \varepsilon_w$$

$$\varepsilon_w \sim N(0, I)$$

$$\text{OLS: } \hat{\beta} = (X_w^T X_w)^{-1} X_w^T y_w$$

$$= (X^T W X)^{-1} X^T W y$$

$$\hat{y}_w = X_w \hat{\beta} = \underbrace{W^{\frac{1}{2}} X (X^T W X)^{-1} X^T W^{\frac{1}{2}}}_{\text{mat matrix}} \cdot W^{\frac{1}{2}} y$$